



EXPLOITS UNIVERSITY

BSc INFORMATION COMMUNICATION AND TECHNOLOGY

BICT 3202 · OBJECT-ORIENTED ANALYSIS AND DESIGN

WEEK 10

Assignment: Object-Oriented Design

PROGRAMME	BSc Information Communication and Technology
COURSE CODE / YEAR	BICT 3202 · Year 3
LECTURER	Francis Fweta — ICT Lecturer
DELIVERY	2h Lecture · 1h Tutorial · 1h Practical · 10h Independent Learning

Prepared by Francis Fweta · Exploits University · Week 10 of 16



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ASSIGNMENT INSTRUCTIONS

- Answer ALL questions in Sections A, B and C.
- Read the Week 10 module and your lecture notes before attempting each question.
- Use the university student registration case study for your examples.
- Diagrams (class, sequence, state) must be neat, complete and correctly labelled.
- Cite the OMG UML 2.5.1 specification and IBM design/analysis model guidance where relevant.
- Submit by the date given by your lecturer. Total: 50 marks.

Section A — Short-Answer Questions (20 marks)

Question A1

Define software design and object-oriented design. [3 marks]

Question A2

Differentiate object-oriented analysis from object-oriented design, with an example. [3 marks]

Question A3

State four goals of good object-oriented design. [2 marks]

Question A4

Define cohesion and coupling. [2 marks]

Question A5

Explain why good design generally seeks high cohesion and low coupling. [2 marks]

Question A6

Differentiate abstraction from encapsulation using a BankAccount example. [3 marks]

Question A7

Explain the Object, Dynamic and Behavioural models. [3 marks]

Question A8

What is traceability, and why is it valuable? [2 marks]

Section B — Application Questions (20 marks)

Question B1 (12 marks)

Exploits University wants an Online Student Registration System. Students will log in, search for courses, register for courses, drop courses and view registered courses.

- (a) Identify five possible classes for the system. [2 marks]

- (b) Assign at least two responsibilities to each class. [3 marks]
- (c) Draw a class diagram showing attributes, operations, associations and multiplicities. [3 marks]
- (d) Explain how the Object, Dynamic and Behavioural models support one another. [4 marks]

Question B2 (8 marks)

Consider the requirement 'a student registers for a course'.

- (a) Draw a sequence diagram for the course registration flow. [4 marks]
- (b) Draw a state diagram for the Registration object. [2 marks]
- (c) Identify one cohesion and one coupling problem that could arise, and suggest a fix. [2 marks]

Section C — Case Study (10 marks)

A bank wants a mobile banking application allowing customers to check balances, transfer money, pay bills and buy airtime. The system must be secure, support many simultaneous users, and allow new payment methods to be added in future.

As an object-oriented designer:

- (a) Identify five classes and their responsibilities. [3 marks]
- (b) Explain how the three models (object, dynamic, behavioural) would be combined. [3 marks]
- (c) Explain how an interface and a design pattern could support adding new payment methods. [4 marks]

Marking Scheme

Section	Content	Marks
A	Short-answer questions (8 questions)	20
B	Application questions (12 + 8)	20
C	Case study — mobile banking system	10
	TOTAL	50

Submission & Academic Integrity

- Submit a typed document (or neatly handwritten, as directed by your lecturer).
- This assignment is individual work. Plagiarism or copying will attract a penalty under university regulations.
- Hand-drawn diagrams must use a ruler and correct UML notation; software-drawn diagrams are welcome.



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- Where you describe cohesion, coupling, abstraction or encapsulation, illustrate with the given scenario rather than definitions only.
 - Cite any sources you consult using the format used in the Week 10 module references.